

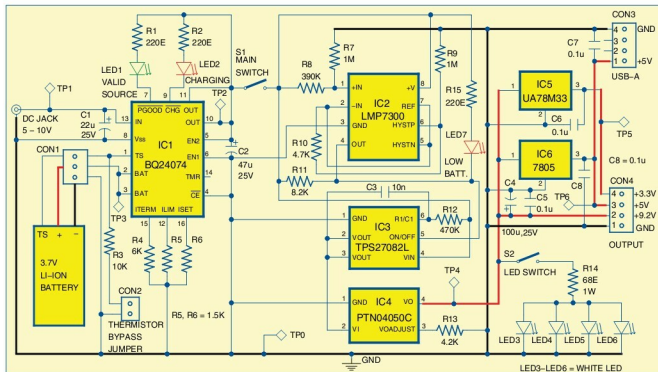


Fig. 3: Circuit diagram of the mini rechargeable power supply

The charging input source is connected to pins 13 (IN) and 8 (V_{SS}) of IC1 through the DC jack. Most of the Li-ion battery packs have at least three pins—positive, negative and NTC thermistor out. The NTC thermistor is built into the battery pack itself to monitor the battery temperature. The three pins of the Li-ion battery are connected to the TS (pin 1), BAT (pins 2 and 3) and ground pin of IC1.

The IC BQ24074 supports only batteries with 10-kilo-ohm thermistors. If your battery does not have this thermistor option, you attach a 10-kilo-ohm from ground to pin 1 (TS) of IC1 by shorting the thermal monitoring jumper to bypass the thermal monitoring feature.

As shown in the circuit (Fig. 3), LED1 is connected to pin 7 (PGOOD) which glows if the input charging source is valid and acceptable for charging the battery. LED2 is connected at pin 9 (CHG) which glows to indicate that the battery is in charging mode. BQ24074 IC works on linear topology, so a 22µF (C1) electrolytic capacitor at the input and a 47µF (C2) at the output is required for stability purpose.

Since 1200mAh battery is used in this power supply project, let us go through the designing steps for BQ24074 IC. First, for setting the charging current at 0.5C, we have to find the value of R_{set} that will be connected at pin 16 (I_{set}) to ground. For 1200mAh of battery capacity, 1C=1200mA and thus 0.5C=600mA. So, for I_{set}=0.5C=600mA, the formula for calculating R_{set} (as per datasheet of BQ24074) is:

$$R_{set} = (K_{set} / I_{set})$$

$$\text{Value of } K_{set} = 890$$

$$\text{Thus, } R_{set} = (890 / 600\text{mA}) = 1.48\text{-kilo-ohm}$$

The value of 1.48-kilo-ohm is not a standard value, so we choose the nearest standard value of 1.5-kilo-ohm. It will set the charging current I_{set} to about 593mA, which is acceptable for the purpose. Thus R_{set} =1.5-kilo-ohm (which is resistor R6 in the circuit).

For setting the charge termination current at 0.1C=120mA, we find out the value of R_{term} that is connected from pin 15 (I_{term}) to ground. The formula for calculating R_{term} (as per datasheet of BQ24074) is:

$$R_{term} = (I_{term} \times R_{set}) / 0.03$$

$$\text{Thus } R_{term} = (120\text{mA} \times 1.5\text{k}) /$$

Test Points

Test point	Details
TP0	0V
TP1	Charging source voltage
TP2	4.4V if charging source is connected
TP3	Battery voltage
TP4	9.2V
TP5	3.3V
TP6	5V

0.03=6-kilo-ohm (which is resistor R4 in the circuit).

Now let us calculate the R_{lim} value at pin 12 of IC1 that will set the maximum input current consumption limit for IC1, which includes the battery charging current plus the current supplied to the load. We select a nominal value of about 1A for the purpose in our project.

$$\text{So, for } I_{lim} = 1\text{A}$$

$$R_{lim} = (K_{lim} / I_{lim})$$

$$K_{lim} = 1610$$

$$\text{Thus, } R_{lim} = (1610 / 1\text{A}) = 1.610\text{-kilo-ohm}$$

Here, 1.61-kilo-ohm is not a standard value, so we choose the nearest standard value of 1.5-kilo-ohm. It will give us I_{lim} =1.073A, which is accept-